

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

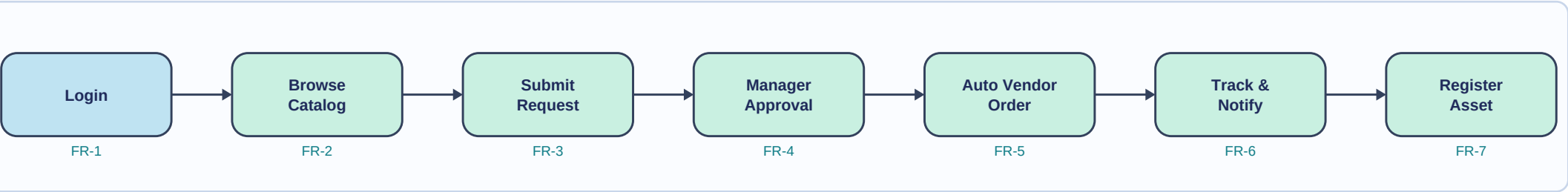
Date	01 October 2026
Team ID	SWTID-2026-3408
Project Name	Streaming IT Procurement: Automating Standard Laptop Orders
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Roles	- Login through company SSO - Login through email and password - Role-based access (Employee, Manager, IT Admin)
FR-2	Laptop Catalog	- View approved standard laptop models with specs and price - IT Admin adds, updates or removes models
FR-3	Laptop Order Request	- Submit request with model, quantity and delivery location - Confirmation via email with request ID - Edit or cancel request before approval
FR-4	Approval Workflow	- Notify manager of pending requests - Approve or reject with comments - Automatic budget-limit check
FR-5	Automated Vendor Ordering	- Auto-generate purchase order for approved requests - Send purchase order to vendor via API / email - Store order and invoice records
FR-6	Order Tracking & Notifications	- Real-time order status on dashboard - Email / in-app notifications at each stage - Alerts for delivery delays
FR-7	Asset Registration & Reports	- Confirm delivery and register laptop asset tag - Reports on orders, spending and delivery delays

Solution Flow (Functional Requirements)



Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, responsive portal so an employee can place a standard laptop order in under 3 steps, on desktop or mobile, with clear status messages.
NFR-2	Security	Company SSO, role-based access control, encrypted data in transit (HTTPS/TLS) and at rest, and an audit log of every approval and order action.
NFR-3	Reliability	Orders and approvals must not be lost or duplicated; failed vendor API calls are retried automatically and logged for the IT team.
NFR-4	Performance	Pages load within 3 seconds and order submission and approval actions respond within 2 seconds under normal load.
NFR-5	Availability	System available 99.5% of the time during business hours, with scheduled maintenance announced in advance and daily data backups.
NFR-6	Scalability	Handles growth in employees and order volume (e.g. bulk orders during onboarding) by scaling cloud resources without redesign.